

Capstone Project Proposal: Marvel vs. DC Movie Success

Problem Identification

Problem Statement:

The superhero movie industry has become a major part of the entertainment business, with Marvel and DC producing some of the most recognisable and commercially successful films. However, movie studios, producers, and audiences may have different perceptions of what makes a superhero movie successful.

The goal of this project is to analyze Marvel and DC live-action theatrical movies to identify the factors associated with movie success and determine whether there are meaningful differences between the two franchises. Then analysis will examine factors such as IMDB ratings, Rotten Tomatoes critic and audience scores, movie genre, runtime, release year, and box-office performance.

The project will ultimately use data science techniques to determine which factors are most strongly associated with movie success and whether one franchise consistently outperforms the other based on measurable criteria.

Context:

The entertainment industry makes significant investments in producing and marketing movies, making it important to understand the factors associated with audience reception and financial performance. Marvel and DC provide an interesting comparison because both franchises have produced movies across multiple decades while developing different characters, genres, directors, and approaches to storytelling.

This project will use historical movie data to identify trends and relationships between movie characteristics, audience and critic ratings, and box-office performance. The analysis will combine exploratory data analysis with predictive modeling to determine whether movie success can be estimated from information available about a film.

Context

The project will be considered successful if the analysis:

- Identifies meaningful differences in performance between Marvel and DC movies.
- Determines which movie characteristics are most strongly associated with financial and critical success.

- Develops a machine learning model capable of predicting a defined measure of movie success with reasonable performance.
- Evaluates model performance using appropriate metrics and a test dataset.
- Produces clear visualizations and business-oriented insights that can be understood by a non-technical audience.

For this project, movie success may be measured using box-office gross, IMDB rating, or a defined success classification. The final target variable will be selected after further exploration of the dataset.

Scope of the Solution Space:

The project will focus specifically on Marvel and DC live-action movies released theatrically. The analysis will include data preparation, exploratory data analysis, feature engineering, statistical analysis, and machine learning.

Potential techniques include:

- Descriptive statistics and data visualization
- Correlation analysis
- Regression modeling to predict box-office performance
- Classification modeling to predict whether a movie meets a defined success threshold
- Feature importance analysis
- Model comparison and evaluation

The project will focus on identifying patterns and making predictions rather than attempting to determine the overall artistic quality of a movie.

Constraints

Several limitations may affect the analysis. The dataset covers movies through 2020, so more recent releases will not be included. The dataset also contains information collected from external sources, meaning some variables may contain missing values or inconsistencies.

Box-office performance can also be influenced by factors that are not included in the dataset, such as marketing budgets, competition from other movies, release timing, economic conditions, inflation, streaming availability, and audience demographics.

Additionally, correlation between a movie characteristic and its success will not necessarily mean that one factor directly causes the other.

Stakeholders

The primary stakeholder for this project would be a movie studio or entertainment executive interested in understanding factors associated with superhero movie performance.

Other potential stakeholders include:

- Movie producers
- Marketing and distribution teams

- Financial analysts
- Entertainment industry researchers
- Movie audiences and fans

The results could help stakeholders better understand historical trends, evaluate factors associated with movie performance, and support data-informed decisions about future movie projects.

Data Sources

The primary dataset will be the Marvel vs. DC - IMDB & Rotten Tomatoes dataset available through Kaggle.

The dataset contains Marvel and DC live-action theatrical movies released between 1944 and 2020. It includes variables such as:

- Movie title
- Release year
- Genre
- Runtime
- MPA rating
- IMDB rating
- IMDB votes
- Box-office gross
- Director
- Leading actors
- IMDB description
- Rotten Tomatoes critic consensus
- Rotten Tomatoes critic score
- Rotten Tomatoes audience score
- Marvel/DC designation

The dataset was compiled using information from sources including Wikipedia, IMDB, and Rotten Tomatoes

Dataset: MARVEL vs. DC - IMDB & Rotten Tomatoes on Kaggle

Proposed Approach and Deliverables

The project will begin by acquiring and examining the dataset to understand its structure, data types, missing values, and potential data-quality issues. The data will then be cleaned and prepared for analysis.

Next, exploratory data analysis will be performed to compare Marvel and DC movies across ratings, box-office performance, genres, release years, and other relevant characteristics. Features will then be selected or engineered for use in predictive models.

Several machine learning approaches will be evaluated, potentially including linear regression, logistic regression, decision trees, and random forest models. Model performance will be evaluated using appropriate and compared against baseline performance.

The final results will be used to develop data-driven conclusions about factors associated with superhero movie success.

The project deliverables will include:

1. GitHub Repository containing the complete project work and code.
2. Project Report documenting the problem, methodology, analysis, modeling, results, and conclusion.
3. Slide Deck presenting the project, key findings, visualizations, model performance, and recommendations.
4. Final Analysis explaining which factors were most associated with movie success and how Marvel and DC compared based on the available data.